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1. Ranking from slowest to fastest in term of rate of growth.

Answer:

$$O(1) \rightarrow O(\log n) \rightarrow O(n) \rightarrow O(n \log n) \rightarrow O(n^5)$$

2. Find the Big-O complexity by looking at the dominant term.

a) $4n^2 + 20n + 9$

$\therefore$ Dominant term is $4n^2$ that is $O(n^2)$

b) $25n^3 + 15$

Dominant term is $25n^3$ that is $O(n^3)$

c) $n^3 + 5 \log n + 6$

Dominant term is n^3 that is $O(n^3)$ is greater than $O(n \log n)$

d) $10^5 + 105n + 10^5 n^2$

Dominant term is $10^5 n^2$ that is $10^5 n^2$ is greater than $O(n)$

e) $72n(n + 3n^2)$

From: $72n(n + 3n^2)$
 $= 72n^2 + 216n^3$

$\therefore 216n^3$ is the dominant term that is $O(n^3)$ is greater than $O(n^2)$.

2. f) $2^8 + 324n - 5$

The dominant term is $324n$ that is $O(n)$ is greater than $O(1)$

3. Answer

worst case we use $O(n)$ because we are looping.

```
int total (int sum) {
```

```
    sum = 0;
```

```
    for (int i = 1; i <= n; i++)
```

```
        sum = sum + i;
```

```
}
```

```
return sum;
```

4. a)

The complexity happens at looping where the it runs number of iterations.

That is $O(n)$ at for $i := 0$ to number do:

```
    x := x + 1
```

```
end for.
```


4. b) Complexity in term of cycles which happens when choice = 2 hence followed by the loop

$\therefore$ The complexity $O(N)$.

c) The complexity happens on both inner and outer loops perform (iter) number of iterations. by $O(1)$

$$\therefore (\text{Iter}) \times (\text{iter}) = (\text{iter}^2) = n^2$$

$\therefore$ The complexity is $O(n^2)$ or $O(\text{iter}^2)$.

d) The complexity doesn't change as they both use loop with a (iter) number of iterations.

Note: This ~~que~~ loops deals only with natural number

$\therefore$ The complexity is $O(\text{iter}^2)$

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5. i) $f(n) \leq c \cdot g(n)$.

ii) $f(n) \geq c \cdot g(n)$

6. Calculation and answer

show that $4n+12$ is $O(n)$

$$c > 0 \text{ and } k \geq 0$$

$$\therefore c = 5 \text{ and } k = 12$$

$$\therefore 4n+12 \leq 5n$$

$\therefore$ with constant $k = 12$ it holds $\forall n \geq 12$

$$\therefore 4(12)+12 \leq 5(12)$$

$$60 \leq 60$$

Conclusion $4n+12$ is $O(n)$ because it holds equality at boundary

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7. Show that $5n^2 - 16$ is $O(n^2)$

Solution.

$$5n^2 - 16$$

$$\text{let } c = 5$$

$\rightarrow$

$$5n^2 - 16 \leq 5n^2$$

$$-16 \leq 0. (\text{True})$$

Conclusively: $5n^2 - 16 \leq 5n^2$ $n \geq 1$ That makes $5n^2 - 16$ is $O(n^2)$